Homework Cover Page Format

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Acknowledgement: None

1.

Known: $k_g = 1.5\ W/m\cdot K$, $L_g = 3mm$, $k_a = 1\ W/m\cdot K$, $L_a = .1mm$

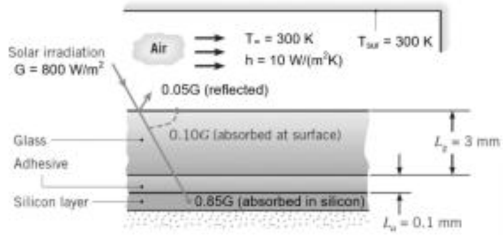
$\eta = a - bT_{si}$, $a = .553$, $b = .001 K^{-1}$, $R_c = 10^{-4} m^2\cdot K/W$

$G = 800\ W/m^2$, 5%-10%-85% per layer, $T_\infty = 300K$, $h = 10\ W/m^2\cdot K$, $h_{rad} = 5\ W/m^2\cdot K$

Find:

a) Resistor network, b) $T_{g,top}$ & T_{si}, c) η & P

Schematic:

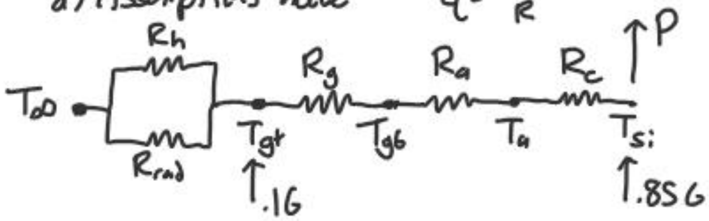


Properties: None

Assumptions: Si layer of 0 thickness, 1-D heat transfer, no contact resistance between glass & adhesive, steady state

Analysis:

a) Assumptions above $q = \frac{\Delta T}{R}$



$R_h = \frac{1}{h}$, $R_{rad} = \frac{1}{h_{rad}}$, $R_g = \frac{L_g}{k_g}$, $R_a = \frac{L_a}{k_a}$, $R_c = R_c$, $P = .85G\eta$

b) $(T_{gt} - T_\infty)\left(\frac{1}{R_h} + \frac{1}{R_{rad}}\right) = .1G + .85G(1-\eta)$

$$T_{gt} = T_\infty + \frac{G(.95 - .85(a - bT_{si}))}{h + h_{rad}}$$

$(T_{si} - T_{gt}) = .85G(1-\eta)(R_g + R_a + R_c)$

$$T_{gt} = T_{si} - .85G(1 - (a - bT_{si}))\left(\frac{L_g}{k_g} + \frac{L_a}{k_a} + R_c\right)$$

$325.6K + .6453\, T_{si} = -.83K + 1.0015\, T_{si}$

$\Rightarrow T_{si} = \underline{342.3\ K}$

$\Rightarrow T_g = \underline{341.1\ K}$

c) $\eta = a - bT_{si} = .2107$

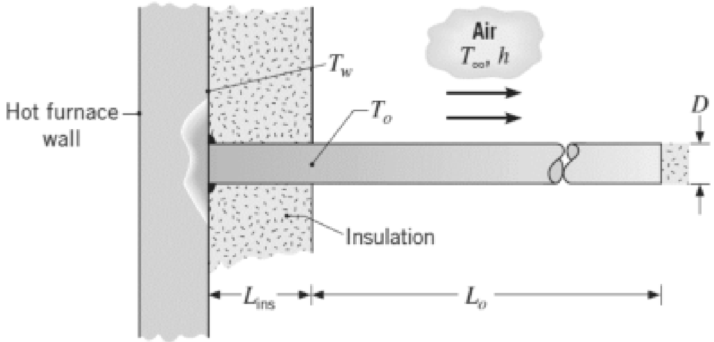
$\Rightarrow P = \eta \cdot .85G = \underline{143.3 W}$

2.

Known: $D = 10\,mm$, $k = 80\,W/m\cdot K$, $T_w = 200°C$
$L_{ins} = 200\,mm$, $T_{max} = 100°C$, $T_\infty = 20°C$
$h = 20\,W/m^2\cdot K$,

Find: a) T_o, b) T_o for $L_o = 200\,mm$, c) min L_o

Schematic:



Properties: None

Assumptions: Insulated tip, no radiation, steady state

Analysis:

a) Adiabatic tip condition: $q = M\tanh(mL)$

$$m^2 = \frac{hP}{kA_c} = \frac{h\,\pi D}{k\,\frac{1}{4}\pi D^2} = \frac{4h}{kD}$$

$$M = \sqrt{hPkA}\,\theta_b = \sqrt{hk\tfrac{1}{4}\pi^2 D^3}\,(T_o - T_\infty)$$

$$q = (T_w - T_o)\cdot\frac{k}{L}\cdot\frac{1}{4}\pi D^2 = \frac{\pi}{2}\sqrt{hkD^3}(T_o - T_\infty)\tanh(mL)$$

$$\boxed{T_o = \frac{\sqrt{kD}\,T_w + 2L_i\sqrt{h}\tanh\left(\sqrt{\frac{4h}{kD}}L_o\right)T_\infty}{\sqrt{kD} + 2L_i\sqrt{h}\tanh\left(\sqrt{\frac{4h}{kD}}L_o\right)}}$$

b) $T_o = \underline{81.5\,C}$ Meets the limit

c) $$\tanh^{-1}\left(\frac{\sqrt{kD}(T_o - T_w)}{2L_i\sqrt{h}(T_\infty - T_o)}\right)\cdot\sqrt{\frac{kD}{4h}} = L_o = .0733\,m$$

$\underline{73.3\,mm}$

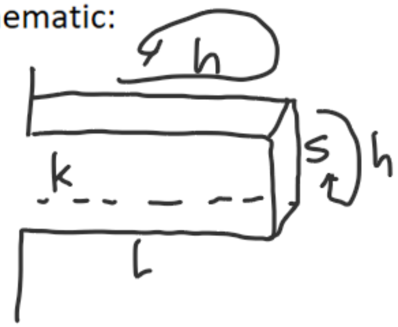
3.

Known: $s = 10\ mm$, $k = 400\ W/m\cdot K$

$h = 100\ W/m^2\cdot K$, $\eta_f = .6$

Find: a) L, b) R_f & ε_f

Schematic:



Properties: None

Assumptions: Steady state

No radiation

Analysis:

$$m = \sqrt{\frac{hP}{kA_c}} = \sqrt{\frac{h\cdot 4s}{ks^2}} = \sqrt{\frac{4h}{ks}} \qquad \frac{M}{\theta_b} = \sqrt{h\cdot 4s\cdot k\cdot s^2} = \sqrt{4hks^3}$$

a) $$\eta_f = \frac{q_f}{q_{max}} = \frac{1}{h(4sL+s^2)} \cdot \sqrt{4hks^3} \cdot \frac{mk\sinh(mL) + h\cosh(mL)}{mk\cosh(mL) + h\sinh(mL)}$$

w/ solver: $L = .149\ m$

b) $$R_f = \frac{\theta_b}{q_f} = \frac{1}{\sqrt{4hks^3}} \cdot \frac{mk\cosh(mL) + h\sinh(mL)}{mk\sinh(mL) + h\cosh(mL)} = 2.754\ K/W$$

$$\varepsilon_f = \frac{q_f}{q_{nf}} = \frac{1}{hs^2} \cdot \sqrt{4hks^3} \cdot \frac{mk\sinh(mL) + h\cosh(mL)}{mk\cosh(mL) + h\sinh(mL)} = 36.31$$